Mohawk Watershed Symposium

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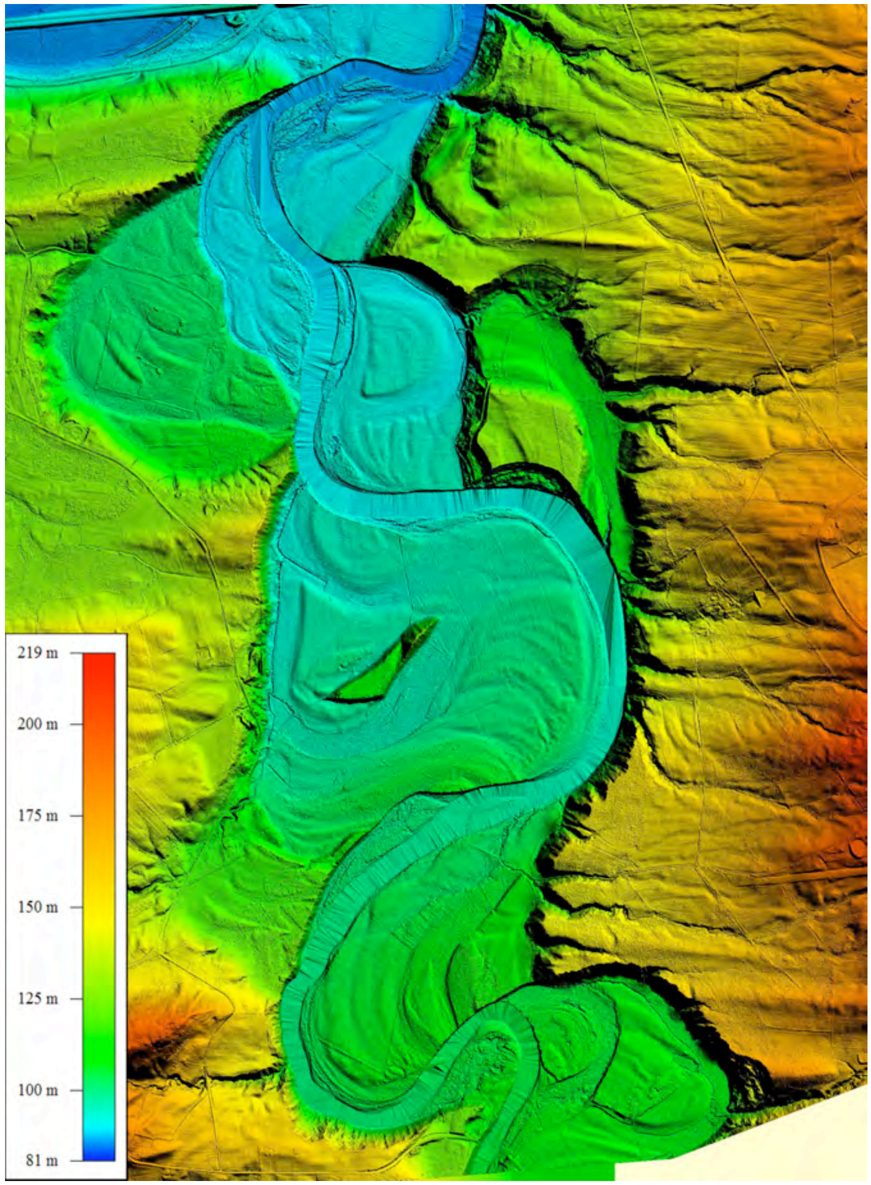

Abstracts and Program

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MAPPING AND VOLUMETRIC CALCULATION OF THE JANUARY 2010 ICE JAM FLOOD, LOWER MOHAWK RIVER, USING LIDAR AND GIS

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Ice jams are an annual occurrence on the Mohawk River (Johnston and Garver, 2001; Lederer and Garver, 2001; Scheller and others, 2002; Garver and Cockburn, 2009). As a northern temperate river, ice jams are expected and the lower Mohawk is particularly vulnerable to jams and the hazards associated with them. Breakup involves ice floes that commonly form ice jams (or dams) that occur when the frozen river breaks up and the moving ice gets stuck due to restriction of flow at channel constrictions and areas of reduced flood plain. Historically we know that the time of ice out and ice jam formation occurs on the rising limb of the hydrograph, when the floodwaters are building. When flow starts to rise it is not uncommon for unimpeded ice runs to develop, but invariably the ice gets blocked or impeded along the way by constrictions in the river, especially where the flood plain is reduced in size.

An important issue in understanding ice jams and where they form is how much water can get backed up behind ice dams that block the flow of the water (see Robichaud and Hicks, 2001; White and others, 2007). It is typical for these features to form, but then break up as water levels increase (Jasek, 1999). In a sense they are self regulating because rising water causes the ice jam to float, which ultimately results in self destruction. When this does occur, there is an ice jam release wave that propagates downstream (Watson et al., 2009). This release of water can itself cause flooding, and it is clearly recorded as an increase in instantaneous discharge downstream. In fact, in many break up floods, the highest instantaneous discharge is in fact a surge that has resulted from the release of an ice jam. The highest instantaneous discharge recorded on the Mohawk River (143k cfs), resulted from just such an event in March 1964.

The mid-winter break up event of 25-26 January 2010 caused significant ice jams to form in the lower part of the Mohawk River. Moderately warm temperatures and heavy rain from a south-to north-tracking Atlantic storm caused considerable melting and rapid increase in discharge on the Mohawk River and its main tributaries, especially Schoharie Creek, which drains the northern Catskills. The highest rainfall amounts were in the headwaters of the Schoharie Creek and were ~5 inches, but elsewhere in the lower basin totals were only about 1 inch. Although rain and melting occurred in the upper parts of the drainage basin, the effects were limited.

January 2010 Ice Jam

Ice accumulation and maximum water levels suggest that the main jam occurred at the Boston and Maine (B&M) rail bridge, which crosses the Mohawk River from Glenville to Rotterdam Junction. The western part of this bridge is essentially on the edge of the SI plant, which had constant monitoring of water levels and video surveillance of the ice. From the video record it is clear that after several minutes of re-adjustment, and a rapid water rise of about 1 foot, the jam released at 09:44 AM on 26 January. Following this release, there was rapid and continuous movement of the ice floe down the Mohawk, and a sharp reduction in water levels. The highest level recorded at the SI plant was 244' at their independent observation station.

Because it appears that the front of the release wave made it from Rotterdam Junction downstream to Cohoes (39 km) by 11:00 AM (26 Jan), this would suggest that the front of the release wave travelled at an average rate of 31.2 km/hr (19.4 mph) over that entire distance. We estimate that the jam first formed at the B&M rail bridge at or before 11:45 PM on 25 January. At this point we consider the volume of water that was backed up in the ice jam, and to do this we calculate the volume of the ice jam release (from the hydrograph downstream) and we use a LiDAR topographic model in the flooded area to estimate the volume of backed up water (Figure 1).

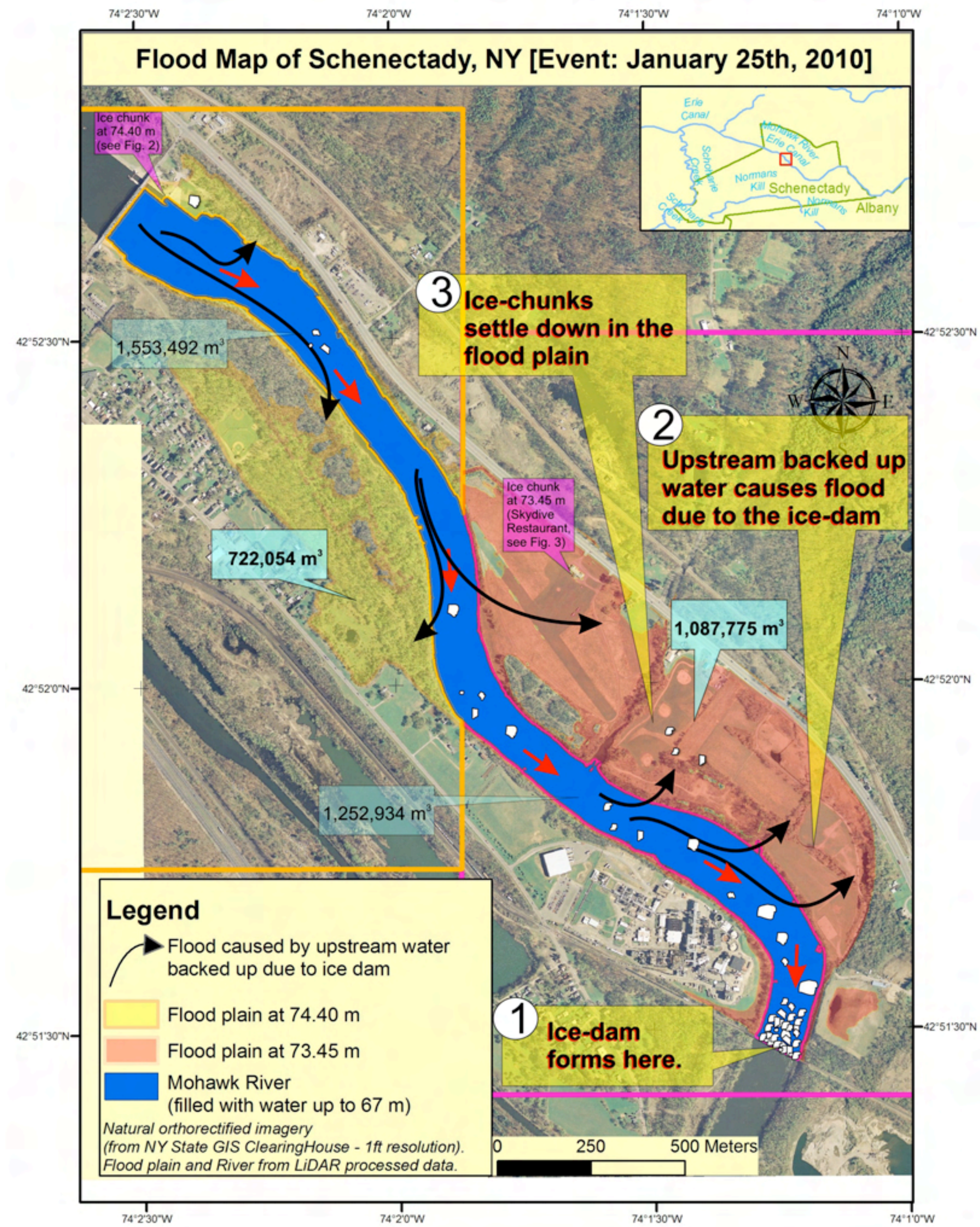

Figure 1: Flood Map where it shows the flood affected area. The flood took place on the 25-26th January (2010) due to the ice-dam that formed in this constricted part of the floodplain. Volumetric calculation results are shown for each sector on the map.

LiDAR volumetric calculation model

Here we test whether flood model applications using LiDAR are successful where topographic relief is low and changes occur gradually. Such digital elevation models (DEM) are particular useful for flood simulation in rural or urban areas. Although important topographic features and properties are not simulated explicitly by Air-LiDAR (such as trees) ground points provide a very realistic digital elevation model of decimeter accuracy. In urban areas, features like roads or buildings have an important effect on flooding and as such must be accounted for in the model set-up.

An accurate calculation of the flood volume requires a digital elevation model of less than 1-meter accuracy. In this study we calculated the volume between the flood plain and the maximum elevation of the ice jam induced flood on January flood from field observations and a LiDAR developed DEM of 0.11 m accuracy. Buildings, affected trees and other existing infrastructure were used to determine the maximum flood elevation.

The flooded study area is located between the New York State Canal System Lock 9 (E9 Lock) and the B&M Rail Bridge at the Schenectady International (SI) Plant (Figure 1). A DEM with grid size of 0.11 m grid was generated from LiDAR data and served as a base line case for various flood simulations. Ideally data processing is supported by a field survey to obtain specific observations and elevation measurements of highest observed water levels (Figure 2). Due to the low gradient of the flood plain, the elevations of the high water mark was estimated in two different target areas. The river area has been delineated using the lowest elevation values, which are essentially bank full conditions (67 m). The lack of information about river levels prior to jamming makes it difficult to fully assess the volumetric calculation, but bank full conditions are a reasonable starting estimate. For this reason, the two target polygon areas were subtracted by the river's polygon (clipped). However, in this study we present the volumetric calculation for both target areas of the river area as separate numbers (Figure 1). The volumetric calculation that took place on the Mohawk River had a 67 m water elevation base line of the river and the flood at 73.45 m and at 74.40 m water elevation, respectively for the two target areas. The main methods that were used to specify the flooded areas were raster to feature process with a prior reclassification of the water values.

The volumetric calculation of the flood has shown that the northwestern portion of the area (near the E9 Lock) was flooded by 722,054 m^3 of water (Figure 1) covering an area of 301,233 m^2. The southeastern portion of the area flooded (from the Skydive restaurant to the Chemical Plant) was flooded by 1,087,775 m^3 covering an area of 525,601 m^2. Assuming that the water level at the river before the ice-dam formation was 67 m then the river was flooded by 2,806,426 m^3. The maximum volume of water that flooded the land derived from the flood of the 24-25th of January caused by the ice-dam between the E9 lock station and the Chemical Plant was 1,809,829 m^3, and it covered an area of 826,834 m^2. Therefore the estimate of the total volume delayed by the ice jam was approximately 4.6 million m^3 using volumetric calculations based on LiDAR-derived topography.

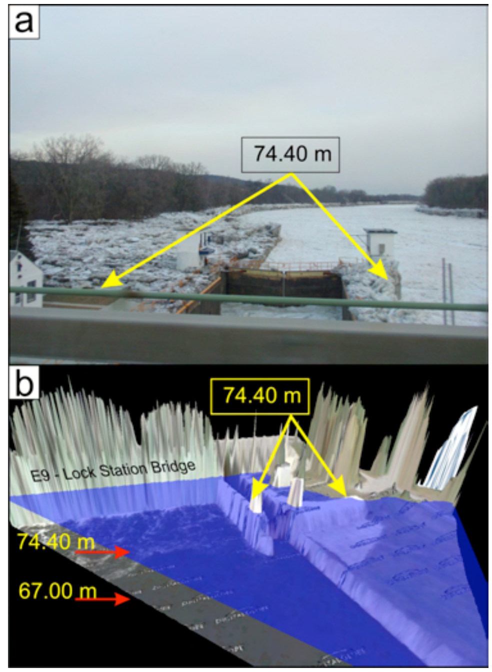

Figure 2: (a) Field observations from the E9 Lock station; (b) water flood model derived from the LiDAR DEM (0.11 m resolution) to determine the accurate flood elevation level.

Hydrograph Separation

A volumetric comparison of the LIDAR based flood volume calculation was conducted using USGS stage and discharge data from the Cohoes Falls station on the Mohawk River (USGS 01357500). Hydrograph separation is a common method used to determine the runoff volume for a given hydrograph component. Graphical separation is the simplest technique and is used extensively in simple runoff events (Singh, 1992). To determine the volume of water released after the jam broke on Jan 26, a straight line was drawn from the time the hydrograph rose rapidly (~11:00am) to intersect with the falling limb of the hydrograph (Figure 3). The slope of this line approximated the slope of the rising limb, prior to jam formation and intersects the falling limb at 3:45pm on Jan 26. It was estimated that 0.0047 km^3 of water (4.7 million m^3) was delayed by the ice jam, by calculating the area between the hydrograph and the straight line. Graphical separation is admittedly simple and has the potential to over or under estimate volumes of flow, but given the data available this method was the most appropriate for the January 26, 2010 ice jam.

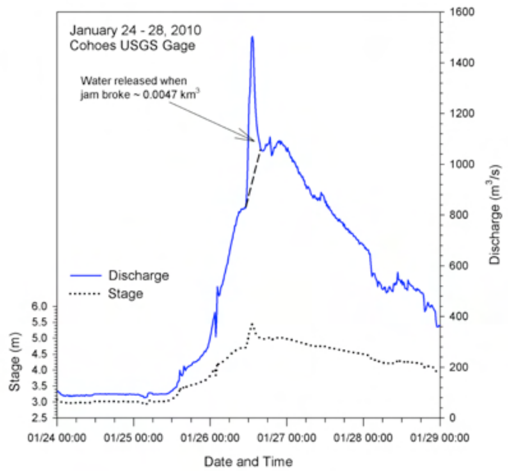

Figure 3. Hydrograph and water level (stage) in the 2010 event (from Cohoes Falls, NY). Estimation of the volume of water that surged through the system is c. 4.7 million m^3.

Conclusions

As ice jams form and break-up there are clearly critical thresholds that are reached that ultimately cause the self-destruction of the ice front. Our calculations of the volume of the flooded area (4.6 million m^3), and the volume of water recorded downstream using hydrographic separation (4.7 million m^3), are, remarkably, in agreement. We suspect continued studies of volume of water behind ice jams in different reaches of the lower Mohawk River will shed light on the critical thresholds for ice build-up and the effects of the release of water downstream (i.e. Brufau, P. and Garcia-Navarro, P., 2000; Nzokou et al., 2009).

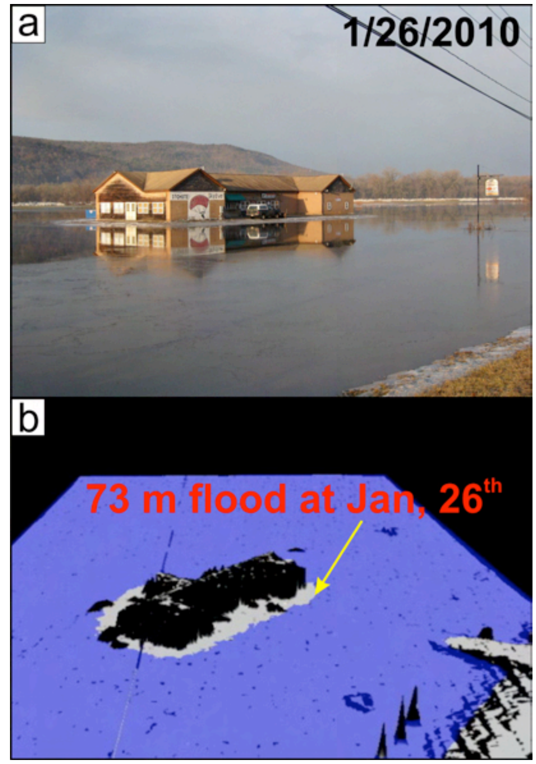

Figure 4: (a) The flooded area as it appears around the Skydive Restaurant in the morning of the 26th of January; (b) LiDAR 3D representation of the flooded area shown by the blue color.

A key piece of data that is required in the future is a real-time monitoring network using pressure transducers that can provide fast reliable data on the condition of the ice movement along several key parts of the river that are prone to ice jamming (Robichaud and Hicks, 2001; White et al., 2007).

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